

sion.⁴ We recommend that patients with a blind painful eye have ongoing assessment, including diagnostic imaging, and that histopathological examination of all enucleation and evisceration specimens be mandatory.

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Intracameral bevacizumab in the treatment of neovascular glaucoma and exudative retinal detachment after brachytherapy in choroidal melanoma

Choroidal melanoma is the most common primary intraocular tumor in adults. Plaque brachytherapy is a common treatment option for medium-sized tumors. Radiation retinopathy and its sequelae, including neovascularization of the iris (NVI), neovascular glaucoma (NVG), and exudative retinal detachment (ERD), are among the known side effects of radiation treatment.¹

NVI can develop as soon as 9 months after radiation (mean time of 27 months) and secondary NVG develops in up to 23% of eyes at 5 years.^{2,3} ERD occurs following plaque radiotherapy in up to 16% of patients.⁴ Treatment options for managing NVG secondary to brachytherapy include medical therapy, panretinal photocoagulation, cyclodestructive procedures, anti-vascular endothelial growth factor (VEGF) therapy, and, in some cases, enucleation. Typically, there are no specific treatments directed to ERDs in treated choroidal melanoma patients.

Multiple reports have shown that intravitreal bevacizumab (Avastin, Genentech Inc, San Francisco, Calif.) an anti-VEGF agent, reduces neovascular activity and vascular permeability in ocular tissues. Bevacizumab can be administered intravitreally or intracamerally to reduce NVI secondary to diabetes and other retinal vascular diseases.^{5,6}

We report the use of intracameral bevacizumab in a patient with NVG and ERD following brachytherapy for a medium-sized choroidal melanoma.

A 45-year-old woman was treated with iodine-125 (I-125) brachytherapy for a medium-sized choroidal melanoma (thickness of 6.8 mm, 10 mm at its largest diameter). There was no retinal detachment (RD) at the time of diagnosis. Fourteen months post treatment, a total ERD was noted despite local tumor control (4.26 mm thickness). Three

months later, she developed a painful left eye and decreased visual acuity to no light perception (NLP). The intraocular pressure (IOP) was 50 mm Hg. Corneal edema was present, with 360° of NVI; the angle was open and neovascularization of the angle was noted. The tumor height was 4.35 mm by ultrasound. Maximum medical therapy was initiated to treat the elevated IOP. Panretinal photocoagulation could not be performed because of the extent of the RD. Bevacizumab (1.25 mg) was injected intracamerally since the intravitreal approach was deemed unsafe in the presence of an RD.

One week post injection, the IOP was 36 mm Hg. On examination, there was regression of the iris and angle neovascularization, which was evident on iris angiography (Fig. 1). One month post injection, visual acuity remained NLP, IOP was 20 mm Hg, and some iris vessels were visible. Intracameral bevacizumab injections were repeated

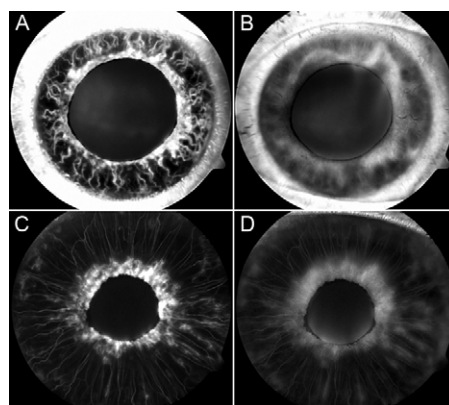


Fig. 1—Extensive iris neovascularization noted in the early (A) and late (B) phase of iris angiography, pre-bevacizumab injection. One week after bevacizumab injection, there is reduction of iris neovascularization, as observed in the early (C) and late (D) phases of the angiogram.